# ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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**Abstract.** Vision–language models (VLMs) such as Vintern-1B bring strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark usually requires fine-tuning on many high-resolution image tiles or training a new architecture. We propose **ViBridge-VQA**, which improves Vintern-1B without updating any of its pre-trained weights. From a single image view, the frozen decoder receives two kinds of visual tokens: *global* tokens, produced from the `[CLS]` vector by a small trainable bridge, and *local* tokens from Vintern-1B's own pre-aligned projector, pooled to a budget chosen as the knee of the accuracy–cost Pareto front. On the reasoning-oriented AutoViVQA benchmark, ViBridge-VQA trains 1.35% of the parameters and reaches 54.6 F1 on a stricter image-disjoint split, comparable to the fine-tuned Vintern-1B while its decoder reads 36 times fewer visual tokens. A lightweight decoder LoRA raises F1 to 58.2. Reusing the pre-trained projector beats learning a new connector, and the gain of the local tokens carries over to two other Vietnamese VQA datasets.

**Keywords:** Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

## 1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. Large vision–language models (VLMs) such as BLIP-2 [13], Flamingo [1] and LLaVA [20] generate free-form answers by connecting a pre-trained vision encoder to a pre-trained language model. For Vietnamese, the open Vintern-1B [8] connects an InternViT-300M encoder [7] to a Qwen2.5-0.5B decoder [27], and benchmarks such as ViVQA [30] and the reasoning-oriented AutoViVQA [22] measure progress in this low-resource setting.

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Adapting such a model to a new benchmark remains expensive. The fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits each test image into six high-resolution tiles plus a thumbnail, so its decoder reads 1,792 visual tokens per image. ViMoE-VQA [21] instead designs a task-specific mixture-of-experts fusion module. A cheaper option keeps the VLM frozen and trains only a small connector, usually of one of two designs. A *global-only bridge* maps the global image vector to a few tokens: it is cheap but loses small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details but must learn a new mapping to the decoder, which is hard with limited Vietnamese data.

Vintern-1B offers a third option. Its projector is already pre-trained to map image regions into the input space of the decoder, so this alignment need not be learned again. The model also discards the global `[CLS]` vector of its encoder, which summarises the whole scene. We therefore ask two questions. With limited labelled data, is reusing the pre-trained alignment more effective than learning a new connector? And how many global and local tokens does the frozen decoder need?

We propose **ViBridge-VQA**, which feeds the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector, average-pooled to a chosen budget and placed in the original image slot, so the decoder reads region features in the representation it was trained on. *Global* tokens, produced from the `[CLS]` vector by a small trainable bridge, add the scene-level information that Vintern-1B ignores. The bridge is the only trained part, so Vintern-1B stays unchanged and can be shared across tasks. As every token adds inference cost, we treat the number of tokens as a budget and select the knee of the accuracy–cost Pareto front. ViBridge-VQA performs on par with the fine-tuned Vintern-1B without updating any of its weights, while its decoder reads 36 times fewer visual tokens. Our contributions are:

- **ViBridge-VQA**, a global–local bridge that combines trainable global tokens with local tokens from the frozen projector of the VLM, optionally with a lightweight decoder LoRA.
- An explicit **accuracy–cost budget** for the number of visual tokens, selected on validation as the knee of the Pareto front.
- An **analysis of where the gains come from**: local tokens matter more than global ones, reusing the projector beats learning a new connector, and the gain of the local tokens carries over to other Vietnamese VQA datasets.

## 2 Related Work

*Vietnamese VQA.* Generative VLMs answer with an autoregressive language model instead of classifying over a fixed answer set [13,1,20]. Vietnamese VQA datasets include ViVQA [30], the synthetic ViCLEVR [32], the book-cover ViOCRVQA [26] and the scene-text ViTextVQA [24]. AutoViVQA [22] contributes

37,077 questions on 19,411 COCO images [18], generated by large language models, filtered by an ensemble of validators and labelled with reasoning types. On the modelling side, BARTPhoBEiT [31] and AViVQA-TranConI [23] combine pre-trained visual and Vietnamese language representations, Vintern-1B [8] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [21] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. AutoViVQA and ViMoE-VQA are earlier work of our group: we use the AutoViVQA data but no component of ViMoE-VQA, and the method, the image-disjoint split and all results are new. These approaches train a new model or fine-tune a VLM on multi-tile inputs, whereas we keep Vintern-1B frozen and use a single image view.

*Connectors.* Training only a connector between a frozen vision encoder and a frozen language model is a common way to build VLMs efficiently, and connectors differ mainly in what they must learn. Frozen [33] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [13] learns queries that compress the patch features: it preserves details but is a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [20,19] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder. Honeybee [4] proposes locality-preserving abstractors, and InternVL [7], the basis of Vintern-1B, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA does not learn it again: it reuses the pre-trained projector and trains only a small global bridge (12.9M parameters) on about 26k task questions.

*Visual token budget.* The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [14] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] show with nested pooled budgets that the required budget depends on the task. Other work reduces the tokens directly: DeCo [36] replaces the Q-Former with adaptive average pooling of the patch tokens, TokenPacker [15] learns a compact projector, LLaVA-PruMerge [29] and FastV [5] prune or merge the tokens of an existing model, and VisionZip [35] keeps those that receive the most attention from `[CLS]`. Like DeCo, we use average pooling, but on the output of an already pre-trained projector with the whole VLM frozen, and we choose the budget with explicit criteria: the knee of the Pareto front [28] and TOPSIS [12]. Parameter-efficient methods such as adapters [10], prefix tuning [16] and LoRA [11] reduce the cost of adaptation from the other side, by training a few additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement to ViBridge-VQA.

## 3 Proposed Method

We formulate VQA as conditional answer generation: given an image $I$ and a Vietnamese question $q$, the model generates a short free-form answer $y =$

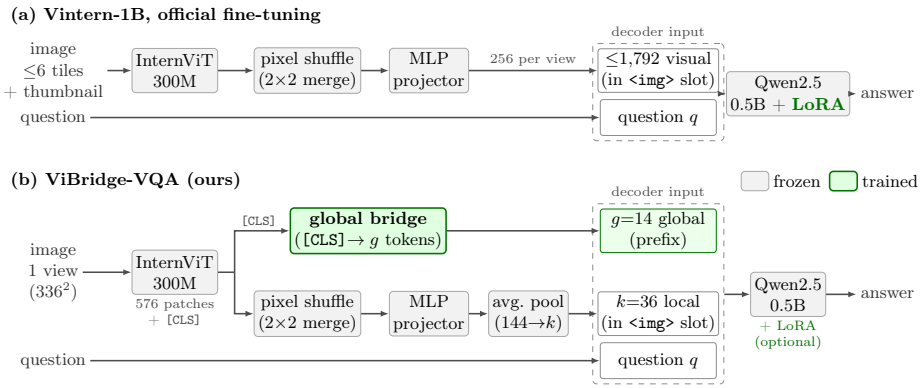



**Fig. 1:** Overview. (a) Official fine-tuning of Vintern-1B on up to seven views. (b) ViBridge-VQA on one view: $k$ local tokens, average-pooled from the 144 region tokens of the frozen projector, fill the image slot, and $g$ global tokens from `[CLS]` precede the prompt. The selected budget is $g = 14$, $k = 36$ (50 tokens). The decoder stays frozen; a LoRA is optional.

$(y_1, \ldots, y_T)$. ViBridge-VQA keeps every pre-trained weight of Vintern-1B [8] frozen and changes only the visual input of its decoder.

### 3.1 Architecture

As shown in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector $P$ and the Qwen2.5-0.5B-Instruct decoder (24 layers, width 896). Each image is resized to one $336 \times 336$ view, i.e., $24 \times 24$ patches of $14 \times 14$ pixels. InternViT, pre-trained at $448 \times 448$, bicubically interpolates its position embeddings to this resolution, which roughly halves the encoder cost (191 instead of 362 GMACs per view). The encoder returns one vector per patch and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: pixel shuffle [6] merges each $2 \times 2$ block of patches into one 4096-dimensional vector, and $P$, pre-trained to align the two backbones, maps it into the decoder input space. This yields a $12 \times 12$ grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

### 3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge $f_\phi$ expands it into $g$ tokens with two linear heads:

$$f_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}, \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head gives an *anchor* token, and the second gives $g - 1$ complementary tokens that present the global

content along different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters (12,857,600 for $g = 14$), and its tokens are placed before the prompt. When the bridge alone is varied from 4 to 18 tokens, F1 peaks at $g = 14$ (Section 4.5), the value used in ViBridge-VQA. Our Global-only model keeps $g = 8$, matching the eight learned queries of the Tile-Attention and light Q-Former bridges it is compared with (Section 4.4). Local tokens are combined with both values, and the pair $(g, k)$ is selected as in Section 3.5.

### 3.3 Local Tokens

To give the decoder region-level detail, ViBridge-VQA also passes local tokens taken directly from the 144 region tokens of $P$. We average-pool the $12 \times 12$ grid to an $s \times s$ grid, so $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each $2 \times 2$ block. The pooled tokens replace the embeddings of $k$ placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`. Because they come from the pre-trained projector rather than a new projection, they stay in the representation that the decoder was trained to read, and they need no parameters.

### 3.4 Training

By default, only the bridge parameters $\phi$ are trained; the encoder, the projector and the decoder stay frozen. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, f_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where $x$ denotes the merged patch vectors. Optionally, ViBridge-VQA is combined with decoder adaptation: a rank-16 LoRA [11] on the query, key, value and output projections of every decoder layer, trained jointly with the bridge, adds 2,162,688 parameters. Percentages of trained parameters are relative to all parameters of the model: the frozen Vintern-1B (about 0.94B, including the encoder, projector, embeddings and output layer) plus the trained modules. With $g = 14$, the bridge trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

### 3.5 Budget Selection

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair $(g, k)$ as a budget. Among the configurations with local tokens, F1 increases with every added token in our experiments, so all of them lie on the accuracy–token Pareto front, which alone does not single one out. We therefore select its *knee* [28], after which additional tokens bring diminishing returns. With accuracy $a$ the validation F1 and cost $c$ the number of visual tokens, both normalised to $[0, 1]$ over the front, the knee maximises $\hat{a} - \hat{c}$, i.e., it lies furthest above the line joining the cheapest and the most accurate configuration. We check the choice with TOPSIS [12]. Both criteria use validation only: test results are reported for every configuration but never used for selection.

# 4 Experiments

## 4.1 Dataset and Metrics

*Dataset.* We use AutoViVQA [22], built on MS COCO images [18], where each question has five short free-form answers and a reasoning type. Its original 80/20 division is made by question, so 2,941 of the 7,416 evaluation questions (40%) concern images also seen in training. We instead split by image. After removing 273 duplicated (image, question) pairs and 97 questions without a reasoning label, the remaining 36,707 questions are split 70/15/15 by image (seed 42), so no test image is seen in training: 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. This split is stricter: a random question-level split raises the F1 of Global-only by about 0.5 points.

*Metrics.* Following AutoViVQA and ViMoE-VQA, we report Accuracy, word-level Precision, Recall and F1, BLEU [25], ROUGE-L [17], METEOR [2] and CIDEr [34] against the five references. Precision, Recall and F1 compare the word sets of the answer and of the reference with the highest F1, and are averaged over questions. Accuracy is the share of answers identical to a reference, ignoring case; BLEU is corpus-level BLEU-4 on whitespace tokens.

## 4.2 Setup

*Implementation details.* The encoder, projector and decoder come from Vintern-1B-v3.5 [8] and stay frozen. We train with AdamW (learning rate $2 \times 10^{-4}$, effective batch size 8) on the first reference answer. Bridges are trained for two epochs, after which the validation loss and CIDEr stop improving. Models with a decoder LoRA are trained for one epoch, with the bridge (if any) and the LoRA learned jointly from scratch; Global-only + LoRA is also reported after three epochs, its best setting, which Table 6 and Fig. 3 use. Decoding is greedy with at most 50 new tokens, and every model fits on one 16 GB NVIDIA T4. Each configuration is trained with three seeds (42, 123, 3407), four for Global-only (adding 2026), and tables report mean ± std.

*Baselines.* Table 1 compares with two groups of models: (1) fine-tuned on AutoViVQA: ViT5_ViT, BARTPhoBEiT [31], Vintern-1B and ViMoE-VQA [21], and (2) zero-shot: Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5. Their results are taken from the AutoViVQA and ViMoE-VQA papers. The fine-tuned Vintern-1B follows the official recipe: six $448 \times 448$ tiles plus a thumbnail per test image and a rank-16 LoRA on the language model for one epoch. The analysis also uses *Global-only*, the bridge of Eq. (1) with $g = 8$ and no local tokens, with and without a decoder LoRA. Our models are *ViBridge-VQA (k=36)*, with $g = 14$, and *ViBridge-VQA (k=36) + LoRA*.

**Table 1:** Results on AutoViVQA. Baselines are as reported in AutoViVQA [22] and ViMoE-VQA [21] on the original split; ours are mean ± std over three seeds on the stricter image-disjoint test split. Best per column in bold. All values ×100.

| | Model | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|---|
| Fine-tuned | ViT5_ViT | 7.97 | 46.84 | 50.33 | 48.52 | 4.13 | 46.89 | 31.02 | 72.68 |
| | BARTPhoBEiT | 8.81 | 45.30 | 46.48 | 45.88 | 4.33 | 44.83 | 24.57 | **188.96** |
| | Vintern-1B | 13.01 | 52.47 | 55.12 | 53.76 | 6.11 | 51.93 | 35.25 | 72.84 |
| | ViMoE-VQA | 9.55 | **62.89** | 58.60 | **60.69** | 12.54 | 47.12 | 39.10 | 88.76 |
| Zero-shot | Vintern-1B | 0.12 | 17.52 | 19.87 | 17.55 | 1.91 | 25.84 | 23.93 | 8.54 |
| | LLaMA 3.2 | 0.36 | 23.96 | 73.71 | 36.16 | 3.62 | 36.11 | 30.01 | 62.84 |
| | Gemini 2.0 F. | 0.55 | 27.20 | 74.10 | 39.79 | 4.41 | 39.60 | 31.72 | 74.42 |
| | Gemini 2.5 F. | 0.22 | 24.43 | **76.66** | 24.75 | 0.39 | 37.27 | 31.22 | 71.90 |
| | GPT-5 | 10.84 | 47.20 | 55.20 | 50.89 | 6.07 | 47.30 | 33.34 | 84.20 |
| Ours | ViBridge-VQA ($k$=36) | 11.75 ±0.21 | 55.36 ±0.21 | 56.56 ±0.21 | 54.56 ±0.19 | 18.52 ±0.18 | 52.64 ±0.19 | 45.53 ±0.33 | 110.58 ±0.95 |
| | ViBridge-VQA ($k$=36) + LoRA | **14.67** ±0.30 | 59.02 ±0.12 | 59.96 ±0.33 | 58.17 ±0.19 | **22.53** ±0.11 | **56.25** ±0.22 | **49.32** ±0.37 | 122.88 ±0.93 |

**Table 2:** Sources of visual tokens with a frozen decoder, on validation. Params: trained parameters. Best trained model in bold. All values ×100.

| Visual tokens from | Tokens | Params | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|---|---|
| Residual | 1 | 4.86M | 5.63 ±.59 | 46.31 ±.34 | 48.79 ±.33 | 46.02 ±.36 | 12.98 ±.15 | 44.23 ±.38 | 36.86 ±.31 | 86.48 ±.77 |
| Global-only | 8 | 7.35M | 9.21 ±.05 | 51.54 ±.15 | 52.48 ±.21 | 50.74 ±.17 | 16.88 ±.23 | 49.03 ±.12 | 41.38 ±.21 | 99.07 ±.47 |
| Tile-Attention | 8 | 4.14M | 4.99 ±1.10 | 45.49 ±1.10 | 48.05 ±.60 | 45.14 ±.92 | 12.34 ±.32 | 43.28 ±.95 | 36.11 ±.73 | 84.14 ±1.40 |
| Light Q-Former | 8 | 27.6M | 6.94 ±.32 | 47.74 ±.80 | 49.18 ±.51 | 47.06 ±.66 | 13.99 ±.33 | 45.37 ±.63 | 37.57 ±.60 | 88.07 ±1.89 |
| Full Q-Former | 16 | 69.4M | 6.97 ±.33 | 47.42 ±.70 | 49.02 ±.35 | 46.84 ±.47 | 13.76 ±.36 | 45.10 ±.41 | 37.43 ±.34 | 87.74 ±1.24 |
| C-Abstractor | 9 | 19.87M | 8.04 ±.07 | 49.41 ±.32 | 50.93 ±.32 | 48.86 ±.33 | 15.27 ±.13 | 47.17 ±.33 | 39.60 ±.33 | 94.05 ±1.32 |
| Patch-pool | 8 | 0.92M | 5.58 ±.27 | 46.11 ±.39 | 48.52 ±.49 | 45.78 ±.44 | 12.71 ±.49 | 43.99 ±.47 | 36.69 ±.46 | 85.83 ±1.43 |
| Local tokens only, untrained | 36 | 0 | 0.00 | 18.07 | 68.17 | 27.55 | 2.47 | 20.82 | 34.50 | 67.20 |
| **ViBridge-VQA** ($k$=36) | 50 | 12.86M | **12.38** ±.22 | **55.86** ±.21 | **56.72** ±.33 | **54.96** ±.28 | **18.73** ±.23 | **53.10** ±.27 | **46.01** ±.42 | **113.29** ±1.01 |

### 4.3 Main Results

*ViBridge-VQA is on par with the fine-tuned Vintern-1B.* Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 (zero-shot) to 54.56 (Table 1). This is comparable to the fine-tuned Vintern-1B (53.76), although none of our test images is seen in training, while the decoder reads 50 instead of 1,792 visual tokens and only 1.35% of the parameters are trained. Only Accuracy is lower (11.75 versus 13.01). On test, the bridge alone with the same 14 global tokens reaches 51.02 F1 (±0.10), so the 36 local tokens add 3.54 F1.

*LoRA adds 3.6 F1.* With a decoder LoRA, ViBridge-VQA reaches 58.17 F1 (+3.61) with 1.57% of the parameters trained, and gives the best Accuracy (14.67), BLEU (22.53), ROUGE-L (56.25) and METEOR (49.32). ViMoE-VQA, which trains its fusion module and decoder on AutoViVQA, keeps the best Precision and F1. Results are stable: on test, the std is at most 0.19 for F1 and 0.95 for CIDEr, and test F1 is within 0.4 of validation F1.

### 4.4 Token Sources

Table 2 compares three sources of visual tokens. *Residual* (one token) and *Global-only* (Eq. (1), $g = 8$) read the `[CLS]` vector. Five bridges learn to compress the 577 encoder tokens: *Tile-Attention* (self-attention and eight learned queries), the light and full Q-Former [13] (7 and 15 queries, two and four layers; the full one

**Table 3:** Effect of the token budget, on validation. Time: minutes to generate all validation answers on one T4. Selected budget in bold. All values ×100.

| $g$ | $k$ | Tokens | Time | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 0 | 8 | 33.9 | 9.21 ±.05 | 51.54 ±.15 | 52.48 ±.21 | 50.74 ±.17 | 16.88 ±.23 | 49.03 ±.12 | 41.38 ±.21 | 99.07 ±.47 |
| 14 | 0 | 14 | 37.2 | 9.74 ±.21 | 52.47 ±.11 | 53.16 ±.14 | 51.56 ±.08 | 17.76 ±.01 | 49.86 ±.05 | 42.14 ±.17 | 101.18 ±.52 |
| 8 | 1 | 9 | 34.2 | 9.33 ±.29 | 52.03 ±.12 | 52.75 ±.10 | 51.03 ±.12 | 16.54 ±.27 | 49.21 ±.13 | 41.37 ±.11 | 99.27 ±.58 |
| 8 | 9 | 17 | 35.5 | 10.25 ±.26 | 53.63 ±.04 | 54.72 ±.16 | 52.82 ±.11 | 17.58 ±.03 | 50.96 ±.05 | 43.56 ±.20 | 106.11 ±.55 |
| 8 | 36 | 44 | 36.0 | 12.15 ±.44 | 55.19 ±.43 | 56.14 ±.40 | 54.32 ±.41 | 18.22 ±.39 | 52.48 ±.43 | 45.31 ±.42 | 111.88 ±1.43 |
| 8 | 144 | 152 | 41.7 | 13.83 ±.12 | 56.94 ±.20 | 58.10 ±.15 | 56.15 ±.08 | 19.82 ±.00 | 54.21 ±.06 | 47.51 ±.05 | 119.05 ±.25 |
| 14 | 1 | 15 | 35.3 | 9.67 ±.14 | 52.50 ±.40 | 53.48 ±.14 | 51.67 ±.25 | 17.30 ±.36 | 49.93 ±.24 | 42.36 ±.19 | 101.66 ±1.11 |
| 14 | 9 | 23 | 34.7 | 10.55 ±.22 | 53.96 ±.10 | 54.66 ±.11 | 52.98 ±.08 | 17.83 ±.23 | 51.17 ±.10 | 43.71 ±.13 | 106.02 ±.80 |
| **14** | **36** | **50** | **35.8** | **12.38** ±.22 | **55.86** ±.21 | **56.72** ±.33 | **54.96** ±.28 | **18.73** ±.23 | **53.10** ±.27 | **46.01** ±.42 | **113.29** ±1.01 |
| 14 | 144 | 158 | 41.1 | 14.19 ±.36 | 57.36 ±.07 | 58.58 ±.03 | 56.61 ±.04 | 20.18 ±.13 | 54.70 ±.05 | 48.00 ±.05 | 120.71 ±.12 |

also reads the question), C-Abstractor [4] (convolution blocks around a $3 \times 3$ pooling, 9 tokens instead of the 144 or more of Honeybee) and *Patch-pool* (a linear projection averaged into eight bands). All are randomly initialised, as a new connector would be for a low-resource task, and trained with the same recipe. The third source is our local tokens, alone or with global tokens (ViBridge-VQA).

*Reusing the projector works best.* The learned compressors stay 1.9–5.6 F1 below Global-only, despite up to 9.4× more parameters. Patch-pool, a minimal baseline that averages bands of encoder tokens after a newly learned projection, reaches 45.78 F1; it differs from our local tokens in the pooling (1D bands instead of a 2D grid) and in the projection. In contrast, adding 36 or 144 local tokens to Global-only ($g = 8$) raises F1 by 3.58 and 5.41 (Table 3). The projector also wins at a similar number of tokens: 17 tokens ($g = 8$, $k = 9$) give 52.82 F1 against 46.84 for the 16-token full Q-Former, and 9 tokens ($g = 8$, $k = 1$) give 51.03 against 48.86 for the 9-token C-Abstractor. Distilling the projector into Global-only does not help: a cosine loss between the mean tokens of the bridge and of the projector gives 50.35 F1, and a KL loss between their answer distributions 50.46, both below the 50.74 obtained without distillation. Without any trained part, the 36 local tokens give only 27.55 F1, as the frozen model answers in long sentences (24.8 words on average).

### 4.5 Token Budget

*Local tokens matter more than global tokens.* Table 3 and Fig. 2b vary $g \in \{8, 14\}$ and $k \in \{1, 9, 36, 144\}$. At $g = 14$, going from 1 to 144 local tokens adds 4.94 F1 and 19.05 CIDEr, whereas going from 8 to 14 global tokens adds only 0.2–0.8 F1 at a fixed $k$. The gain per token falls from 0.07 F1 (23 to 50 tokens) to 0.015 F1 (50 to 158 tokens).

*The knee is $g = 14$, $k = 36$.* Since F1 grows with every added token, all configurations with local tokens lie on the accuracy–token Pareto front. Its knee (Section 3.5, Fig. 2b) is $g = 14$, $k = 36$ for every metric, and also with generation time as the cost; TOPSIS and a post-hoc knee on test agree. $k = 144$ is 1.65 F1 higher, but $k = 36$ obtains 72% of its gain over Global-only (67% over the 14-token bridge of Table 4) with 32% of its tokens.

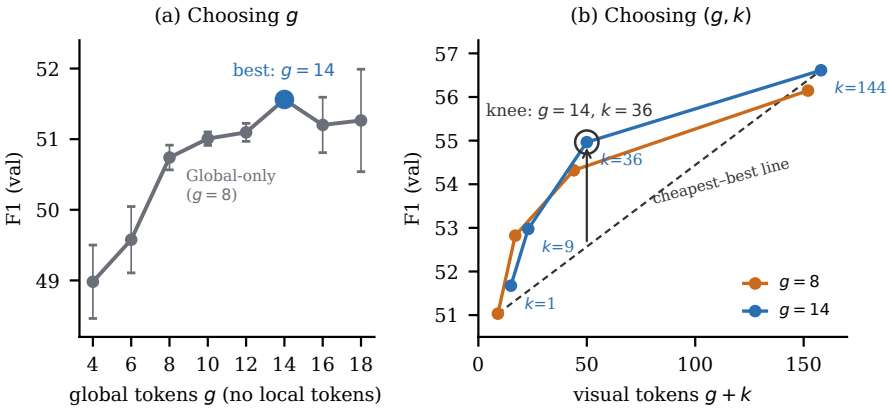



**Fig. 2:** Budget selection (validation F1). (a) Global-only peaks at $g = 14$. (b) With local tokens, the knee is the configuration furthest above the line joining the cheapest and the most accurate one.

**Table 4:** Global-only with $g$ tokens, on validation. Selected $g$ in bold. All values $\times 100$.

| $g$ | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|
| 4 | 7.77 ±.27 | 49.57 ±.48 | 51.07 ±.57 | 48.98 ±.52 | 15.17 ±.48 | 47.23 ±.58 | 39.50 ±.59 | 93.56 ±1.42 |
| 6 | 8.38 ±.39 | 50.34 ±.53 | 51.43 ±.43 | 49.58 ±.47 | 15.67 ±.29 | 47.91 ±.48 | 40.09 ±.50 | 95.51 ±1.59 |
| 8 | 9.21 ±.05 | 51.54 ±.15 | 52.48 ±.21 | 50.74 ±.17 | 16.88 ±.23 | 49.03 ±.12 | 41.38 ±.21 | 99.07 ±.47 |
| 10 | 9.24 ±.14 | 51.95 ±.06 | 52.58 ±.13 | 51.01 ±.10 | 17.03 ±.22 | 49.26 ±.11 | 41.46 ±.07 | 99.20 ±.25 |
| 12 | 9.41 ±.08 | 52.03 ±.12 | 52.64 ±.21 | 51.10 ±.13 | 17.06 ±.30 | 49.47 ±.08 | 41.52 ±.23 | 99.26 ±.54 |
| **14** | **9.74** ±.21 | **52.47** ±.11 | **53.16** ±.14 | **51.56** ±.08 | **17.76** ±.01 | **49.86** ±.05 | **42.14** ±.17 | **101.18** ±.52 |
| 16 | 9.44 ±.37 | 51.99 ±.58 | 52.91 ±.22 | 51.20 ±.39 | 17.29 ±.63 | 49.53 ±.36 | 41.88 ±.31 | 100.29 ±.95 |
| 18 | 9.45 ±.45 | 52.13 ±.72 | 52.88 ±.69 | 51.26 ±.73 | 17.41 ±.54 | 49.56 ±.68 | 41.81 ±.69 | 100.46 ±2.02 |

*Global tokens peak at $g = 14$.* Table 4 and Fig. 2a vary $g$ from 4 to 18 for Global-only. Validation F1 rises from 48.98 ($g = 4$) to 50.74 ($g = 8$) and 51.56 ($g = 14$), then falls to 51.20 and 51.26 at $g = 16$ and 18, and CIDEr follows the same trend. We therefore combine the local tokens with $g = 14$, and keep $g = 8$ as the Global-only reference.

*One view cuts the encoder cost 13 times.* The decoder reads 50 visual tokens instead of 1,792 for the official tiling, and the encoder processes one $336 \times 336$ view (191 GMACs) instead of seven $448 \times 448$ views (362 GMACs each). Within our budgets, $k = 144$ makes generation about 15% slower than $k = 36$.

### 4.6 Decoder Adaptation

*LoRA gains more with local tokens.* ViBridge-VQA + LoRA is the best model on every metric of Table 5 (58.17 F1, 122.88 CIDEr), and LoRA gains more on ViBridge-VQA than on Global-only for all eight metrics (+3.61 versus +2.62 F1, +12.31 versus +8.24 CIDEr). Without local tokens, LoRA lifts Global-only

**Table 5:** Effect of a rank-16 decoder LoRA (2.16M parameters, one joint epoch), on test. Bridge parameters: 7.35M for Global-only, 12.86M for ViBridge-VQA. Best in bold. All values ×100.

| Model | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|
| Global-only | 8.86 ±.37 | 51.19 ±.20 | 52.37 ±.21 | 50.48 ±.22 | 16.24 ±.43 | 48.70 ±.19 | 41.10 ±.29 | 96.20 ±1.10 |
| Global-only + LoRA | 10.47 ±.22 | 53.95 ±.04 | 54.72 ±.07 | 53.10 ±.05 | 19.00 ±.16 | 51.29 ±.08 | 43.63 ±.16 | 104.44 ±.45 |
| ViBridge-VQA | 11.75 ±.21 | 55.36 ±.21 | 56.56 ±.21 | 54.56 ±.19 | 18.52 ±.18 | 52.64 ±.19 | 45.53 ±.33 | 110.58 ±.95 |
| **ViBridge-VQA + LoRA** | **14.67** ±.30 | **59.02** ±.12 | **59.96** ±.33 | **58.17** ±.19 | **22.53** ±.11 | **56.25** ±.22 | **49.32** ±.37 | **122.88** ±.93 |

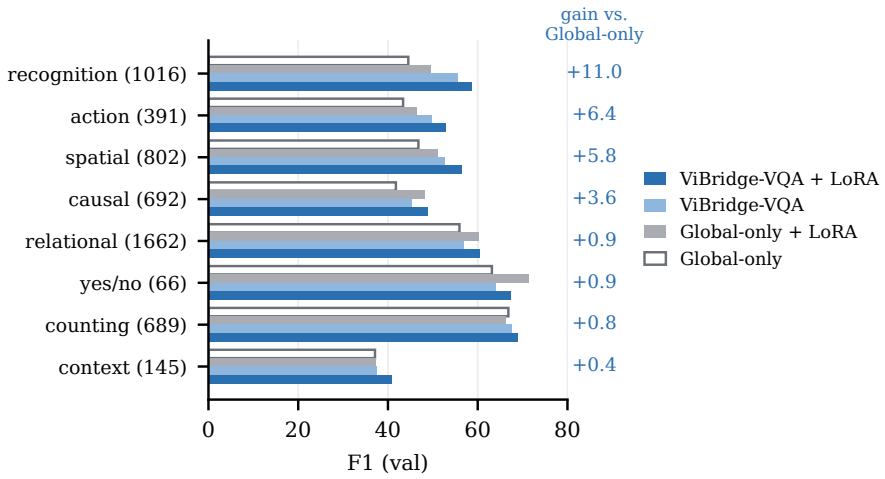



**Fig. 3:** Validation F1 by reasoning type (number of questions), sorted by the gain of ViBridge-VQA over Global-only (right column).

only to 53.10 F1 (54.52 after three epochs), below ViBridge-VQA with a frozen decoder. With LoRA, Residual, Tile-Attention and the two Q-Formers reach only 52.69–53.29 validation F1. With a frozen decoder, the global bridge is essential: it lifts the local tokens from 27.55 to 54.96 validation F1.

### 4.7 Analysis

*Local tokens help most on visible content.* Fig. 3 breaks down validation F1 by reasoning type. ViBridge-VQA adds 11.0 F1 over Global-only on recognition, 6.4 on action and 5.8 on spatial questions, and beats Global-only + LoRA on these types without adapting the decoder. Decoder adaptation helps more on causal and relational questions, which have the longest reference answers (5.0 and 5.6 words), and on the 66 yes/no questions. ViBridge-VQA + LoRA is the best of the four models on seven of the eight types, all but yes/no.

*More exact answers, fewer misses.* On validation, answers are as long as the references (4.3 words on average), and most of them (77–81%) partially overlap the closest reference. Compared with Global-only, ViBridge-VQA raises exact




**Fig. 4:** Validation examples. GT: one of the five references; Ours: ViBridge-VQA ($k$=36); Ours+LoRA: ViBridge-VQA ($k$=36) + LoRA.

**Table 6:** Cross-dataset results (mean ± std over three samples of about 1,000 test questions, 510 for ViVQA; one training seed). Best per dataset in bold. All values ×100.

| Model | Accuracy | Precision | Recall | F1 | BLEU | ROUGE-L | METEOR | CIDEr |
|---|---|---|---|---|---|---|---|---|
| *ViVQA* | | | | | | | | |
| Vintern-1B, zero-shot | 0.00 ±.00 | 16.01 ±.18 | **60.80** ±1.17 | 23.90 ±.24 | 0.27 ±.02 | 15.78 ±.25 | 17.84 ±.19 | 78.63 ±2.07 |
| Global-only + LoRA | 7.38 ±.53 | 31.67 ±.97 | 34.01 ±1.58 | 31.48 ±1.15 | 0.25 ±.09 | 31.48 ±1.16 | 21.92 ±.82 | 84.94 ±2.87 |
| **ViBridge-VQA** | 16.59 ±1.48 | 40.59 ±1.37 | 48.86 ±1.89 | 42.68 ±1.52 | 0.37 ±.00 | 42.65 ±1.55 | 34.62 ±1.37 | 171.21 ±6.06 |
| **ViBridge-VQA** + LoRA | **18.93** ±1.37 | **43.08** ±.93 | 50.88 ±1.71 | **44.76** ±1.21 | **0.63** ±.02 | **44.74** ±1.22 | **36.98** ±1.28 | **183.60** ±7.17 |
| *ViVQA-X* | | | | | | | | |
| Vintern-1B, zero-shot | 0.20 ±.08 | 9.13 ±.35 | **61.15** ±.80 | 14.52 ±.47 | 0.06 ±.01 | 8.98 ±.32 | 11.96 ±.41 | 50.13 ±2.83 |
| Global-only + LoRA | 2.57 ±.09 | 13.19 ±.49 | 26.80 ±1.24 | 15.89 ±.65 | 0.18 ±.02 | 11.94 ±.49 | 9.08 ±.57 | 42.12 ±2.92 |
| **ViBridge-VQA** | 7.40 ±.28 | 20.68 ±.16 | 41.32 ±.26 | 24.91 ±.14 | **0.34** ±.12 | 19.38 ±.25 | 16.00 ±.21 | 79.51 ±1.15 |
| **ViBridge-VQA** + LoRA | **8.11** ±.33 | **23.07** ±.18 | 44.57 ±.06 | **27.64** ±.11 | 0.33 ±.05 | **21.98** ±.13 | **19.24** ±.14 | **95.29** ±1.51 |

matches (F1 = 1) from 9.7% to 13.3% and lowers answers with no word in common with any reference (F1 = 0) from 9.6% to 8.0%; with LoRA, these reach 15.7% and 6.8%.

*LoRA makes answers precise.* In Fig. 4, ViBridge-VQA already finds the relevant content, such as the driver's phone or the helmets, but its answers are vague. The decoder LoRA makes them precise: the driver is using *the phone*, and the helmets are worn *to protect their heads*.

### 4.8 Cross-Dataset Generalisation

Table 6 evaluates the models on ViVQA-X [9] and ViVQA [30], two Vietnamese VQA datasets on COCO images not used for training. About 10% of the sampled questions concern images of our training split, with different questions. Even without LoRA, ViBridge-VQA outperforms Global-only + LoRA and the zero-shot Vintern-1B (F1 24.91 versus 15.89 and 14.52 on ViVQA-X; 42.68 versus 31.48 and 23.90 on ViVQA), and LoRA raises its F1 to 27.64 and 44.76. Zero-shot Vintern-1B keeps the highest Recall only because it answers in long sentences, which also gives it near-zero Accuracy.

## 5 Conclusion

We introduced **ViBridge-VQA**, which improves the Vietnamese VLM Vintern-1B for generative VQA without updating any of its pre-trained weights. For a VLM whose projector is already aligned with its decoder, our results support three findings. First, reusing this projector with a few pooled local tokens is more effective than learning a new connector from limited data. Second, the number of visual tokens can be chosen with an explicit accuracy–cost criterion. Third, local tokens complement decoder adaptation rather than replace it: ViBridge-VQA with a decoder LoRA gives the best result.

Our study has limitations. We train on a single dataset, AutoViVQA, where the task-specific ViMoE-VQA still reaches a higher F1. Without decoder adaptation, the exact-match accuracy of ViBridge-VQA stays below that of the fine-tuned Vintern-1B, and accuracy has not saturated at the 144 local tokens available at our resolution.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many. The code and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

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